

Geographic Analysis

Do enrollment trends cluster geographically, and what regional patterns emerge?

```
In [660... import numpy as np
import pandas as pd
import os
import matplotlib.pyplot as plt
import plotly.express as px
import plotly.graph_objects as go
import seaborn as sns
from scipy import stats
from matplotlib.colors import TwoSlopeNorm
```

```
In [661... enroll_df = pd.read_csv(os.path.dirname(os.getcwd()) + '/data/cleaned_enroll')
```

```
In [662... enroll_df.rename(columns={'State or jurisdiction': 'State'}, inplace=True)
enroll_df.head()
```

```
Out[662... State Year Total_Pub_Under 4y_Pub_Under 2y_Pub_Under Total_Pub_Postb
```

	State	Year	Total_Pub_Under	4y_Pub_Under	2y_Pub_Under	Total_Pub_Postb
0	Alabama	2012-13	216535	130260	86275	345
1	Alabama	2013-14	213669	129045	84624	346
2	Alabama	2014-15	212458	129916	82542	345
3	Alabama	2015-16	212968	131503	81465	344
4	Alabama	2016-17	216115	135080	81035	345

```
In [663... states = [
    "Alabama", "Alaska", "Arizona", "Arkansas", "California", "Colorado", "C
    "Delaware", "Florida", "Georgia", "Hawaii", "Idaho", "Illinois", "Indiar
    "Kansas", "Kentucky", "Louisiana", "Maine", "Maryland", "Massachusetts",
    "Minnesota", "Mississippi", "Missouri", "Montana", "Nebraska", "Nevada",
    "New Jersey", "New Mexico", "New York", "North Carolina", "North Dakota",
    "Oklahoma", "Oregon", "Pennsylvania", "Rhode Island", "South Carolina",
    "Tennessee", "Texas", "Utah", "Vermont", "Virginia", "Washington", "West
    "Wisconsin", "Wyoming", "District of Columbia"
]

state_enrollment = enroll_df[enroll_df["State"].isin(states)]
state_enrollment['State'].unique()
```

```
Out[663...] array(['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'California',
      'Colorado', 'Connecticut', 'Delaware', 'District of Columbia',
      'Florida', 'Georgia', 'Hawaii', 'Idaho', 'Illinois', 'Indiana',
      'Iowa', 'Kansas', 'Kentucky', 'Louisiana', 'Maine', 'Maryland',
      'Massachusetts', 'Michigan', 'Minnesota', 'Mississippi',
      'Missouri', 'Montana', 'Nebraska', 'Nevada', 'New Hampshire',
      'New Jersey', 'New Mexico', 'New York', 'North Carolina',
      'North Dakota', 'Ohio', 'Oklahoma', 'Oregon', 'Pennsylvania',
      'Rhode Island', 'South Carolina', 'South Dakota', 'Tennessee',
      'Texas', 'Utah', 'Vermont', 'Virginia', 'Washington',
      'West Virginia', 'Wisconsin', 'Wyoming'], dtype=object)
```

```
In [664...] state_enrollment = state_enrollment.copy()
state_enrollment['Total_Enrollment'] = state_enrollment['Total_Pub_Under'] +
state_enrollment.head()
```

```
Out[664...]
   State  Year  Total_Pub_Under  4y_Pub_Under  2y_Pub_Under  Total_Pub_Postba
```

0	Alabama	2012-13	216535	130260	86275	346070
1	Alabama	2013-14	213669	129045	84624	346070
2	Alabama	2014-15	212458	129916	82542	346070
3	Alabama	2015-16	212968	131503	81465	346070
4	Alabama	2016-17	216115	135080	81035	346070

```
In [665...] northeast = [
    "Connecticut", "Maine", "Massachusetts", "New Hampshire",
    "New Jersey", "New York", "Pennsylvania", "Rhode Island", "Vermont"
]

midwest = [
    "Illinois", "Indiana", "Iowa", "Kansas", "Michigan", "Minnesota",
    "Missouri", "Nebraska", "North Dakota", "Ohio", "South Dakota",
    "Wisconsin"
]

south = [
    "Alabama", "Arkansas", "Delaware", "Florida", "Georgia", "Kentucky",
    "Louisiana", "Maryland", "Mississippi", "North Carolina", "Oklahoma",
    "South Carolina", "Tennessee", "Texas", "Virginia", "West Virginia"
]

west = [
    "Alaska", "Arizona", "California", "Colorado", "Hawaii", "Idaho",
    "Montana", "Nevada", "New Mexico", "Oregon", "Utah", "Washington",
    "Wyoming"
]
```

```
In [666... regions = {
    "Northeast": northeast,
    "Midwest": midwest,
    "South": south,
    "West": west
}
```

```
In [667... northeast_enrollment = state_enrollment[state_enrollment["State"].isin(north
northeast_enrollment.head()
```

```
Out[667... State Year Total_Pub_Under 4y_Pub_Under 2y_Pub_Under Total_Pub_Po
```

63	Connecticut	2012-13	111481	53253	58228
----	-------------	---------	--------	-------	-------

64	Connecticut	2013-14	109851	52874	56977
----	-------------	---------	--------	-------	-------

65	Connecticut	2014-15	108641	53487	55154
----	-------------	---------	--------	-------	-------

66	Connecticut	2015-16	106337	53576	52761
----	-------------	---------	--------	-------	-------

67	Connecticut	2016-17	103564	53016	50548
----	-------------	---------	--------	-------	-------

```
In [668... southern_enrollment = state_enrollment[state_enrollment["State"].isin(south)
southern_enrollment.head()
```

```
Out[668... State Year Total_Pub_Under 4y_Pub_Under 2y_Pub_Under Total_Pub_Postba
```

0	Alabama	2012-13	216535	130260	86275	344
---	---------	---------	--------	--------	-------	-----

1	Alabama	2013-14	213669	129045	84624	346
---	---------	---------	--------	--------	-------	-----

2	Alabama	2014-15	212458	129916	82542	348
---	---------	---------	--------	--------	-------	-----

3	Alabama	2015-16	212968	131503	81465	344
---	---------	---------	--------	--------	-------	-----

4	Alabama	2016-17	216115	135080	81035	349
---	---------	---------	--------	--------	-------	-----

```
In [669... midwest_enrollment = state_enrollment[state_enrollment["State"].isin(midwest)
midwest_enrollment.head()
```

Out [669...

	State	Year	Total_Pub_Under	4y_Pub_Under	2y_Pub_Under	Total_Pub_Postba
144	Illinois	2012-13	507197	148467	358730	496
145	Illinois	2013-14	496801	145231	351570	496
146	Illinois	2014-15	480483	144381	336102	489
147	Illinois	2015-16	459770	143615	316155	493
148	Illinois	2016-17	443681	139508	304173	486

In [670...

```
western_enrollment = state_enrollment[state_enrollment["State"].isin(west)]
western_enrollment.head()
```

Out [670...

	State	Year	Total_Pub_Under	4y_Pub_Under	2y_Pub_Under	Total_Pub_Postba
9	Alaska	2012-13	28034	27276	758	250
10	Alaska	2013-14	29089	27014	2075	250
11	Alaska	2014-15	27588	26456	1132	230
12	Alaska	2015-16	26118	25925	193	220
13	Alaska	2016-17	25146	24958	188	220

sub-question: Are Northeastern states experiencing different trends than Southern/Western states?

In [671...

```
ne_year_trends = northeast_enrollment.groupby("Year").agg({"Total_Enrollment": "sum"})
ne_year_trends.head()
```

Out [671...

	Year	Total_Enrollment
0	2012-13	2877934
1	2013-14	2862350
2	2014-15	2852403
3	2015-16	2813510
4	2016-17	2780728

In [672...

```
s_year_trends = southern_enrollment.groupby("Year").agg({"Total_Enrollment": "sum"})
```

```
s_year_trends.head()
```

Out [672...

	Year	Total_Enrollment
0	2012-13	6133962
1	2013-14	6036847
2	2014-15	5976291
3	2015-16	5906797
4	2016-17	5892357

In [673... w_year_trends = western_enrollment.groupby("Year").agg({"Total_Enrollment":
w_year_trends.head()

Out [673...

	Year	Total_Enrollment
0	2012-13	4611684
1	2013-14	4562435
2	2014-15	4595443
3	2015-16	4552909
4	2016-17	4527540

In [674... mw_year_trends = midwest_enrollment.groupby("Year").agg({"Total_Enrollment":
mw_year_trends.head()

Out [674...

	Year	Total_Enrollment
0	2012-13	4045950
1	2013-14	3951046
2	2014-15	3805782
3	2015-16	3698222
4	2016-17	3604595

In [675... fig = go.Figure()

Northeast
fig.add_trace(go.Scatter(
 x=ne_year_trends['Year'],
 y=ne_year_trends['Total_Enrollment'],
 mode='lines+markers',
 name='Northeast'
))

Southeast
fig.add_trace(go.Scatter(
 x=s_year_trends['Year'],

```

    y=s_year_trends['Total_Enrollment'],
    mode='lines+markers',
    name='Southeast'
))

# West
fig.add_trace(go.Scatter(
    x=w_year_trends['Year'],
    y=w_year_trends['Total_Enrollment'],
    mode='lines+markers',
    name='West'
))

fig.add_trace(go.Scatter(
    x=mw_year_trends['Year'],
    y=mw_year_trends['Total_Enrollment'],
    mode='lines+markers',
    name='Midwest'
))

fig.update_layout(
    title="Total Enrollment by Region",
    xaxis_title="Year",
    yaxis_title="Total Enrollment",
    legend_title="Region"
)

fig.show()

```

From the appearance of this graph it would seem like the trends all mirror each other regardless of region.

```

In [676... ne_baseline = ne_year_trends[ne_year_trends['Year'] == '2012-13']['Total_Enr
s_baseline = s_year_trends[s_year_trends['Year'] == '2012-13']['Total_Enroll
w_baseline = w_year_trends[w_year_trends['Year'] == '2012-13']['Total_Enroll
mw_baseline = mw_year_trends[mw_year_trends['Year'] == '2012-13']['Total_Enr

print(f"Northeast Baseline: {ne_baseline}")
print(f"Southeast Baseline: {s_baseline}")
print(f"West Baseline: {w_baseline}")
print(f"Midwest Baseline: {mw_baseline}")

```

```

Northeast Baseline: 2877934
Southeast Baseline: 6133962
West Baseline: 4611684
Midwest Baseline: 4045950

```

```

In [677... ne_year_trends

```

Out [677]...

	Year	Total_Enrollment
0	2012-13	2877934
1	2013-14	2862350
2	2014-15	2852403
3	2015-16	2813510
4	2016-17	2780728
5	2017-18	2760260
6	2018-19	2727334
7	2019-20	2696130
8	2020-21	2582373
9	2021-22	2505320

```
In [678] ne_slope, ne_intercept, ne_r_value, ne_p_value, ne_std_err = stats.linregress
print(f"Northeast Slope: {ne_slope}")
print(f"Northeast Intercept: {ne_intercept}")
print(f"Northeast R-squared: {ne_r_value**2}")
print(f"Northeast P-value: {ne_p_value}")
print(f"Northeast Standard Error: {ne_std_err}")
```

Northeast Slope: -38628.64242424243
 Northeast Intercept: 2919663.0909090913
 Northeast R-squared: 0.9067226254914476
 Northeast P-value: 2.1520246405552484e-05
 Northeast Standard Error: 4380.416258681408

```
In [679] ne_year_trends['eindex'] = (ne_year_trends['Total_Enrollment'] / ne_baseline)
```

```
In [680] ##Baseline
ne_slope, ne_intercept, ne_r_value, ne_p_value, ne_std_err = stats.linregress
print(f"Northeast Slope: {ne_slope}")
print(f"Northeast Intercept: {ne_intercept}")
print(f"Northeast R-squared: {ne_r_value**2}")
print(f"Northeast P-value: {ne_p_value}")
print(f"Northeast Standard Error: {ne_std_err}")
```

Northeast Slope: -1.3422351737128946
 Northeast Intercept: 101.44996691755583
 Northeast R-squared: 0.9067226254914471
 Northeast P-value: 2.1520246405552907e-05
 Northeast Standard Error: 0.15220697412384793

```
In [681] px.scatter(ne_year_trends, x=np.arange(len(ne_year_trends['Year']).unique()))
```

```
In [682] s_slope, s_intercept, s_r_value, s_p_value, s_std_err = stats.linregress(np.
print(f"South Slope: {s_slope}")
print(f"South Intercept: {s_intercept}")
print(f"South R-squared: {s_r_value**2}")
```

```
print(f"South P-value: {s_p_value}")
print(f"South Standard Error: {s_std_err}")
```

South Slope: -52915.69696969697
 South Intercept: 6116998.036363637
 South R-squared: 0.8244967400029993
 South P-value: 0.00027991586150859156
 South Standard Error: 8631.527326558591

```
In [683... s_year_trends['eindex'] = (s_year_trends['Total_Enrollment']/ s_baseline)*10
```

```
In [684... s_slope, s_intercept, s_r_value, s_p_value, s_std_err = stats.linregress(np.
print(f"South Slope: {s_slope}")
print(f"South Intercept: {s_intercept}")
print(f"South R-squared: {s_r_value**2}")
print(f"South P-value: {s_p_value}")
print(f"South Standard Error: {s_std_err}")
```

South Slope: -0.8626675054344477
 South Intercept: 99.72344198356032
 South R-squared: 0.8244967400030001
 South P-value: 0.0002799158615085862
 South Standard Error: 0.14071700031005355

```
In [685... px.scatter(s_year_trends, x=np.arange(len(s_year_trends['Year'].unique())),
```

```
In [686... w_slope, w_intercept, w_r_value, w_p_value, w_std_err = stats.linregress(np.
print(f"West Slope: {w_slope}")
print(f"West Intercept: {w_intercept}")
print(f"West R-squared: {w_r_value**2}")
print(f"West P-value: {w_p_value}")
print(f"West Standard Error: {w_std_err}")
```

West Slope: -36331.35151515152
 West Intercept: 4656363.981818182
 West R-squared: 0.6514671000917855
 West P-value: 0.004761800512824026
 West Standard Error: 9395.333619167748

```
In [687... w_year_trends['eindex'] = (w_year_trends['Total_Enrollment']/ w_baseline)*10
```

```
In [688... w_slope, w_intercept, w_r_value, w_p_value, w_std_err = stats.linregress(np.
print(f"West Slope: {w_slope}")
print(f"West Intercept: {w_intercept}")
print(f"West R-squared: {w_r_value**2}")
print(f"West P-value: {w_p_value}")
print(f"West Standard Error: {w_std_err}")
```

West Slope: -0.7878109496477106
 West Intercept: 100.96884309111772
 West R-squared: 0.6514671000917853
 West P-value: 0.004761800512824031
 West Standard Error: 0.2037289115899475

```
In [689... px.scatter(w_year_trends, x=np.arange(len(w_year_trends['Year'].unique())),
```



```
In [690... mw_year_trends['eindex'] = (mw_year_trends['Total_Enrollment']/ mw_baseline)
```

```
In [691... mw_slope, mw_intercept, mw_r_value, mw_p_value, mw_std_err = stats.linregres
print(f"West Slope: {mw_slope}")
print(f"West Intercept: {mw_intercept}")
print(f"West R-squared: {mw_r_value**2}")
print(f"West P-value: {mw_p_value}")
print(f"West Standard Error: {mw_std_err}")
```

West Slope: -2.3928844669585416
 West Intercept: 99.27671105448884
 West R-squared: 0.9901667106074328
 West P-value: 2.5666624371901345e-09
 West Standard Error: 0.08430862117946912

```
In [692... px.scatter(mw_year_trends, x=np.arange(len(mw_year_trends['Year']).unique()))
```

```
In [693... p_values = [ne_p_value, s_p_value, w_p_value, mw_p_value]
```

```
for p_value in p_values:
    if p_value < 0.05:
        print("The null hypothesis can be rejected")
    else:
        print("The null hypothesis cannot be rejected")
```

The null hypothesis can be rejected
 The null hypothesis can be rejected
 The null hypothesis can be rejected
 The null hypothesis can be rejected

This test shows that all of the negative trends are statistically significant at the 0.05 level.

```
In [694... ne_changes = northeast_enrollment.groupby('State')['Total_Enrollment'].apply
s_changes = southern_enrollment.groupby('State')['Total_Enrollment'].apply(1
w_changes = western_enrollment.groupby('State')['Total_Enrollment'].apply(1a
m_changes = midwest_enrollment.groupby('State')['Total_Enrollment'].apply(1a

# print(ne_changes)
# print(s_changes)
# print(w_changes)
# print(m_changes)

f_stat, p_value = stats.f_oneway(ne_changes, s_changes, w_changes, m_changes
print(f"ANOVA p-value: {p_value}")
```

ANOVA p-value: 0.24772554498006544

```
In [695... print(f"Northeast Slope: {ne_slope}")
print(f"South Slope: {s_slope}")
print(f"West Slope: {w_slope}")
print(f"Midwest Slope: {mw_slope}")
```

Northeast Slope: -1.3422351737128946

South Slope: -0.8626675054344477

West Slope: -0.7878109496477106

Midwest Slope: -2.3928844669585416

From this we can see that the northeast enrollment has the steepest decline considering their 2012-13 date as a baseline. The south has the next steepest decline. The west has the slowest decline. However if the anova value of 0.25 is to be believed then the trends are not statistically different.

sub-question: Is there spatial autocorrelation in enrollment trends? (Do neighboring states behave similarly?)

In [696... `import json`

```
state_dict = json.load(open('state_neighbors.json'))
```

In [697... `state_trend = state_enrollment[state_enrollment['State'] == 'Alabama'].group`
`state_trend`

Out [697...

	Year	State	Total_Pub_Under	4y_Pub_Under	2y_Pub_Under	Total_Pub_Postb
--	------	-------	-----------------	--------------	--------------	-----------------

0	2012-13	Alabama	216535	130260	86275	34
1	2013-14	Alabama	213669	129045	84624	34
2	2014-15	Alabama	212458	129916	82542	34
3	2015-16	Alabama	212968	131503	81465	34
4	2016-17	Alabama	216115	135080	81035	34
5	2017-18	Alabama	217755	137019	80736	36
6	2018-19	Alabama	217610	137380	80230	37
7	2019-20	Alabama	216326	136404	79922	38
8	2020-21	Alabama	204632	133781	70851	41
9	2021-22	Alabama	207241	131801	75440	42

In [698... `for state, neighbors in state_dict.items():`
`state_trend = state_enrollment[state_enrollment['State'] == state].group`
`state_baseline = state_trend[state_trend['Year'] == '2012-13']['Total_Er`

```

state_trend['eindex'] = (state_trend['Total_Enrollment'] / state_baseline

state_slope, state_intercept, state_r_value, state_p_value, state_std_err
state_dict[state]['slope'] = state_slope
state_dict[state]['intercept'] = state_intercept
state_dict[state]['r_value'] = state_r_value
state_dict[state]['p_value'] = state_p_value
state_dict[state]['std_err'] = state_std_err

if state_p_value < 0.05:
    state_dict[state]['sig'] = True
else:
    state_dict[state]['sig'] = False

```

In [699... state_dict['Alabama']

```

Out[699... {'neighbors': ['Florida', 'Georgia', 'Mississippi', 'Tennessee'],
'slope': np.float64(-0.9769282076054494),
'intercept': np.float64(100.26594080791443),
'r_value': np.float64(-0.8877080240535666),
'p_value': np.float64(0.0006062015452915853),
'std_err': np.float64(0.17913864438786797),
'sig': True}

```

```

In [700... for state, vals in state_dict.items():
    neighbors = vals['neighbors']
    neighbor_slopes = [
        state_dict[n]['slope']
        for n in neighbors
        if state_dict[n]['sig'] is True
    ]
    if len(neighbor_slopes) == 0:
        vals['signed_neighbor_index'] = np.nan
    else:
        neighbor_avg = np.mean(neighbor_slopes)
        vals['signed_neighbor_index'] = vals['slope'] - neighbor_avg

```

In [701... state_dict['Alabama']

```

Out[701... {'neighbors': ['Florida', 'Georgia', 'Mississippi', 'Tennessee'],
'slope': np.float64(-0.9769282076054494),
'intercept': np.float64(100.26594080791443),
'r_value': np.float64(-0.8877080240535666),
'p_value': np.float64(0.0006062015452915853),
'std_err': np.float64(0.17913864438786797),
'sig': True,
'signed_neighbor_index': np.float64(0.2835207852733451)}

```

```

In [702... map_dict = state_dict.copy()
map_dict.pop('Alaska')
map_dict.pop('Hawaii')

```

```
Out[702... {'neighbors': [],  
            'slope': np.float64(-2.7969147532607512),  
            'intercept': np.float64(98.41155812659451),  
            'r_value': np.float64(-0.968237120644961),  
            'p_value': np.float64(4.2855555432460574e-06),  
            'std_err': np.float64(0.2553589883599495),  
            'sig': True,  
            'signed_neighbor_index': nan}
```

```
In [703... import geopandas as gpd
```

```
states = gpd.read_file(  
    "https://raw.githubusercontent.com/PublicaMundi/MappingAPI/master/data/geo-countries.json"  
)
```

```
In [704... df = (  
    pd.DataFrame.from_dict(state_dict, orient='index')  
    .reset_index()  
    .rename(columns={'index': 'state'})  
)  
  
gdf = states.merge(df, left_on='name', right_on='state', how='left').dropna()  
gdf.head()
```

Out [704...

	id	name	density	geometry	state	neighbors	slope	intercept
0	01	Alabama	94.65	POLYGON ((-87.3593 35.00118, -85.60668 34.9847...	Alabama	[Florida, Georgia, Mississippi, Tennessee]	-0.976928	100.265941
2	04	Arizona	57.05	POLYGON ((-109.0425 37.00026, -109.04798 31.33...	Arizona	[California, Colorado, New Mexico, Nevada, Utah]	-2.126646	94.690029
3	05	Arkansas	56.43	POLYGON ((-94.47384 36.50186, -90.15254 36.496...	Arkansas	[Louisiana, Missouri, Mississippi, Oklahoma, T...	-2.131879	100.515335
4	06	California	241.70	POLYGON ((-123.23326 42.00619, -122.37885 42.0...	California	[Arizona, Nevada, Oregon]	-0.668767	103.267797
5	08	Colorado	49.33	POLYGON ((-107.91973 41.00391, -105.72895 40.9...	Colorado	[Arizona, Kansas, Nebraska, New Mexico, Oklaho...	-0.224448	98.312281

In [705...

```

index_series = gdf['signed_neighbor_index'].dropna()
vmin = index_series.min()
vmax = index_series.max()
vcenter = 0.0 # 0 = same as neighbors

norm = TwoSlopeNorm(vmin=vmin, vcenter=vcenter, vmax=vmax)

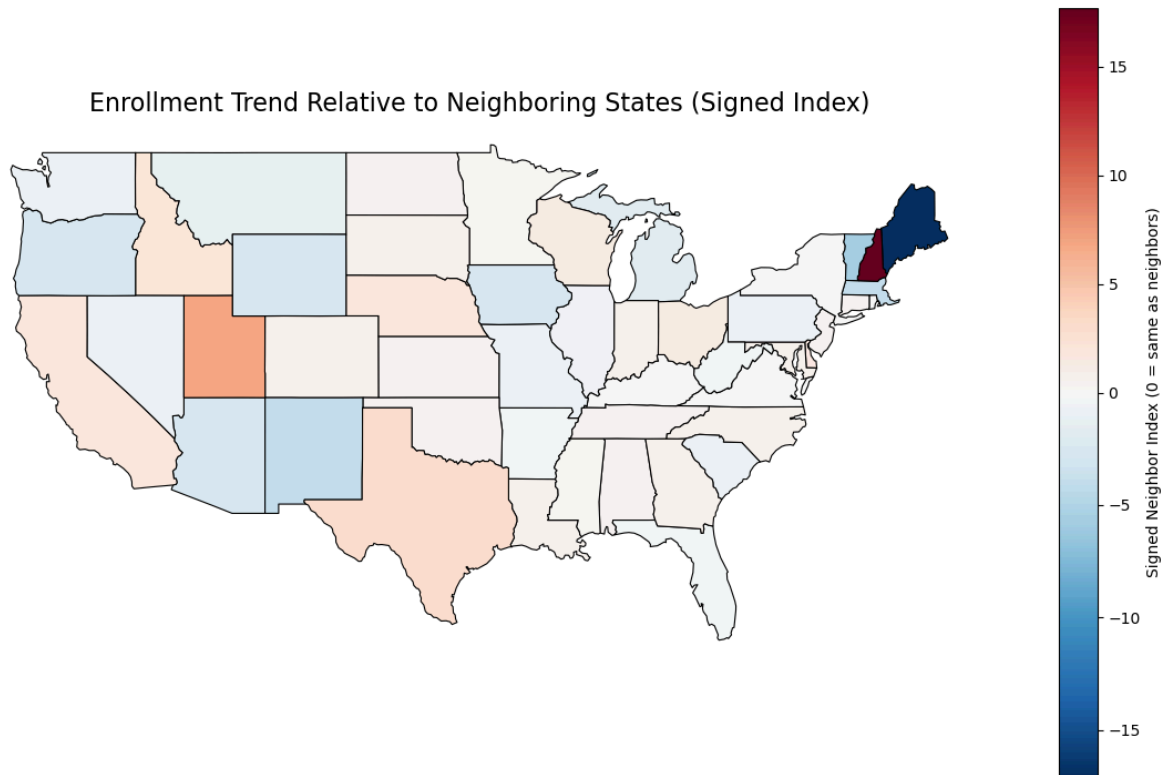
# 6 Plot
fig, ax = plt.subplots(1, 1, figsize=(15, 9))
gdf.plot(
    column='signed_neighbor_index',
    cmap='RdBu_r',
    linewidth=0.8,
    ax=ax,
    edgecolor='black',
    norm=norm,
    missing_kwds={"color": "lightgrey", "label": "No comparison"}
)
ax.set_title("Enrollment Trend Relative to Neighboring States (Signed Index)")
ax.axis('off')

# 7 Colorbar
sm = plt.cm.ScalarMappable(cmap='RdBu_r', norm=norm)

```

```
sm._A = []
cbar = fig.colorbar(sm, ax=ax)
cbar.set_label("Signed Neighbor Index (0 = same as neighbors)")

plt.show()
```



This shows the map of the us with their trends relative to their neighbors. The darker the blue the faster the decline relative to its neighbors.

Statistical test: Morans statistic

```
In [706... from esda.moran import Moran
from libpysal.weights import W
```

```
In [707... # Convert your state_neighbors.json to spatial weights
neighbors = state_dict # Your existing dict

neighbors.pop('Alaska')
neighbors.pop('Hawaii')
# Create weights matrix
w_dict = {state: vals['neighbors'] for state, vals in neighbors.items()}
w = W(w_dict)

w
```

```
Out[707... <libpysal.weights.weights.W at 0x17e1ecd40>
```

```
In [708... state_slopes = [state_dict[state]['slope'] for state in w_dict.keys()]
```

```
In [709... moran = Moran(state_slopes, w)

print(f"Moran's I: {moran.I:.4f}")
print(f"P-value: {moran.p_sim:.4f}")
print(f"Interpretation: {'Significant spatial clustering' if moran.p_sim < 0.05 else 'No spatial clustering'})
```

Moran's I: 0.0329

P-value: 0.1580

Interpretation: No spatial clustering

Question: Is there spatial autocorrelation in enrollment trends? (Do neighboring states behave similarly?)

Answer: All states perform relatively similarly to their neighbors. However as expected the states that are surrounded by faster declining states can appear to be performing better than the reality and then the Moran's I value is not significant further proving it.

sub-question: Are urban vs mixed vs rural states showing different patterns?

For this data we designated a state as Urban if over 70% of the population is Urban, mixed if 40-70% and rural if less than 40%.

```
In [710... urb_rur_df = pd.read_csv(os.path.dirname(os.getcwd()) + '/data/rural_urban_p')
```

```
In [711... urb_rur_df.head()
```

```
Out[711...
      index      0      1      2      3
0  Label (Grouping)  Total:  Urban   Rural  Not defined for this file
1      Alabama  5,024,279  2,900,880  2,123,399                0
2      Alaska   733,391   475,967   257,424                0
3      Arizona  7,151,502  6,385,230   766,272                0
4      Arkansas 3,011,524  1,670,677  1,340,847                0
```

```
In [712... urb_rur_df.columns = ['State', 'Total', 'Urban', 'Rural', 'Not defined for t
urb_rur_df.drop(0, inplace=True)
```

```
In [713... urb_rur_df.head()
```

Out [713...

	State	Total	Urban	Rural	Not defined for this file
--	-------	-------	-------	-------	---------------------------

1	Alabama	5,024,279	2,900,880	2,123,399	0
2	Alaska	733,391	475,967	257,424	0
3	Arizona	7,151,502	6,385,230	766,272	0
4	Arkansas	3,011,524	1,670,677	1,340,847	0
5	California	39,538,223	37,259,490	2,278,733	0

In [714...

```
urb_rur_df['Total'] = urb_rur_df['Total'].str.replace(',', '').astype(int)
urb_rur_df['Urban'] = urb_rur_df['Urban'].str.replace(',', '').astype(int)
urb_rur_df['Rural'] = urb_rur_df['Rural'].str.replace(',', '').astype(int)
```

In [715...

```
r = urb_rur_df['Urban'] / urb_rur_df['Total']
map = r.map(lambda x: 'Urban' if x > 0.7 else 'Rural' if x < 0.4 else 'Mixed')
urb_rur_df['desig'] = map
```

In [716...

```
merged = pd.merge(urb_rur_df, state_enrollment, how='left', on='State')
merged.drop(columns= ['Not defined for this file'], inplace=True)
merged.head()
```

Out [716...

	State	Total	Urban	Rural	desig	Year	Total_Pub_Under	4y_Pub_Uni
--	-------	-------	-------	-------	-------	------	-----------------	------------

0	Alabama	5024279	2900880	2123399	Mixed	2012-13	216535.0	13026
1	Alabama	5024279	2900880	2123399	Mixed	2013-14	213669.0	12904
2	Alabama	5024279	2900880	2123399	Mixed	2014-15	212458.0	12991
3	Alabama	5024279	2900880	2123399	Mixed	2015-16	212968.0	13150
4	Alabama	5024279	2900880	2123399	Mixed	2016-17	216115.0	13508

In [717...

```
rural_df = merged[merged['desig'] == 'Rural']
urban_df = merged[merged['desig'] == 'Urban']
mixed_df = merged[merged['desig'] == 'Mixed']
```

In [718...

```
mixed_df.head()
```


Out [718...

	State	Total	Urban	Rural	desig	Year	Total_Pub_Under	4y_Pub_Un
0	Alabama	5024279	2900880	2123399	Mixed	2012-13	216535.0	13026
1	Alabama	5024279	2900880	2123399	Mixed	2013-14	213669.0	12904
2	Alabama	5024279	2900880	2123399	Mixed	2014-15	212458.0	12991
3	Alabama	5024279	2900880	2123399	Mixed	2015-16	212968.0	13150
4	Alabama	5024279	2900880	2123399	Mixed	2016-17	216115.0	13508

In [719...

```
urban_df = urban_df.drop(510)
```

In [720...

```
urb_changes = urban_df.groupby('State')['Total_Enrollment'].apply(lambda x:
rur_changes = rural_df.groupby('State')['Total_Enrollment'].apply(lambda x:
mix_changes = mixed_df.groupby('State')['Total_Enrollment'].apply(lambda x:

# print(urb_changes)
# print(rur_changes)
# print(mix_changes)

f_stat, p_value = stats.f_oneway(urb_changes, rur_changes, mix_changes)
print(f"ANOVA p-value: {p_value}")
```

ANOVA p-value: 0.9395264975711706

Question: Are urban vs mixed vs rural states showing different patterns?**Answer:** There is no significant difference between rural, mixed and urban states.**sub-question:** Which Census regions/divisions show the strongest declines?

In [721...

```
ne_division = ['Connecticut', 'Maine', 'Massachusetts', 'New Hampshire', 'Rhode Island']
mid_atlantic = ['New Jersey', 'New York', 'Pennsylvania']
e_north_cent = ['Illinois', 'Indiana', 'Michigan', 'Ohio', 'Wisconsin']
w_north_cent = ['Iowa', 'Kansas', 'Minnesota', 'Missouri', 'Nebraska', 'North Dakota', 'South Dakota']
south_atlantic = ['Delaware', 'District of Columbia', 'Florida', 'Georgia', 'Louisiana', 'Maryland', 'North Carolina', 'South Carolina', 'Virginia', 'West Virginia']
east_south_central = ['Alabama', 'Kentucky', 'Mississippi', 'Tennessee']
west_south_central = ['Arkansas', 'Louisiana', 'Oklahoma', 'Texas']
mountain = ['Arizona', 'Colorado', 'Idaho', 'Montana', 'Nevada', 'New Mexico', 'Utah', 'Wyoming']
pacific = ['Alaska', 'California', 'Hawaii', 'Oregon', 'Washington']
```

In [722...

```
divisions = [ne_division, mid_atlantic, e_north_cent, w_north_cent, south_atlantic, east_south_central, west_south_central, mountain, pacific]
```

In [723...

```
div_changes = {}

for i, div in enumerate(divisions):
    # Filter enrollment for the states in this division
    df_div = state_enrollment[state_enrollment['State'].isin(div)]
```

```

# Compute average yearly percent change per state
changes = (
    df_div.groupby('State')['Total_Enrollment']
    .apply(lambda x: x.pct_change().dropna().mean())
    .dropna()
)

# Store results in dictionary
div_changes[f"Division_{i+1}"] = changes

# Optional: ANOVA if you have multiple subgroups within the division
# Example: comparing Urban vs Rural vs Mixed inside this division
#f_stat, p_value = stats.f_oneway(urb_changes, rur_changes, mix_changes)

# print(f"ANOVA p-value: {p_value}") # Only compute if comparing groups

```

```

In [724... f_stat, p_value = stats.f_oneway(div_changes['Division_1'], div_changes['Div
print(f"ANOVA p-value: {p_value}")

```

ANOVA p-value: 0.30222041072658085

There is no significant different in the trends between the 9 regions of census data.
Overall this shows that the country is experiencing an overall decline in enrollment overall.